



SUMMER INTERNSHIP

on

**“AI/ML Engineer Job Readiness and Large Language
Models: A Practical Learning Journey using
NotebookLM during the
BSERC–ISRO Summer Internship Programme”**

Submitted to

**BHARAT SPACE EDUCATION RESEARCH CENTRE
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Submitted by:

Nikhil Singh Chauhan

**M.S – Artificial Intelligence and Cyber Security
Indian Institute of Technology, Patna
Patna, Bihar, India**

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ABSTRACT

The **BSERC–ISRO Def-Space Summer Internship Programme 2026** provided an excellent opportunity to explore emerging technologies and strengthen practical knowledge in the fields of **Artificial Intelligence (AI), Machine Learning (ML), Large Language Models (LLMs), Space Technology, and Defence Applications**. The internship was designed to bridge the gap between academic learning and industry expectations through structured learning modules, technical assignments, and AI-assisted educational activities.

As part of the internship, various assignments were completed to develop technical, analytical, and professional skills. The first assignment focused on evaluating the competencies required for an **AI/ML Engineer** by comparing multiple industry job descriptions with a personal resume using **Google NotebookLM**. This exercise helped identify existing strengths, skill gaps, interview preparation strategies, and a structured learning roadmap for career development.

The second assignment involved an educational study of **Large Language Models (LLMs)**, with particular emphasis on **Llama 3**. Using NotebookLM, technical concepts were explained for different audiences to demonstrate the importance of effective communication in Artificial Intelligence. The activity also encouraged critical thinking by analysing AI-generated explanations and developing improved human-written responses.

Overall, the internship enhanced technical knowledge, problem-solving ability, documentation skills, and the practical application of modern AI tools. It also provided valuable exposure to current technological advancements and reinforced the importance of continuous learning, innovation, and professional development in the rapidly evolving fields of Artificial Intelligence and emerging technologies.

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CHAPTER 1

INTRODUCTION

The rapid advancement of Artificial Intelligence (AI), Machine Learning (ML), Space Technology, and Defence Systems has created a growing demand for professionals equipped with interdisciplinary technical skills and practical knowledge. To bridge the gap between academic learning and industry requirements, internship programmes play a significant role by providing students with opportunities to gain hands-on experience, explore emerging technologies, and develop problem-solving abilities through structured learning and real-world applications.

The **BSERC–ISRO Def-Space Summer Internship Programme 2026** was designed with the objective of introducing students to cutting-edge technologies related to Artificial Intelligence, Space Science, Defence Innovation, Data Analytics, and modern computing systems. The internship provided a comprehensive learning environment through expert sessions, practical assignments, AI-assisted learning tools, and self-paced technical activities. These activities encouraged participants to strengthen their technical foundation while improving analytical thinking, documentation skills, communication abilities, and professional ethics.

During the internship, participants were exposed to various concepts associated with Artificial Intelligence and emerging technologies. Practical assignments required the use of modern AI tools for research, learning, and technical analysis, enabling students to understand how these technologies can be effectively utilized for solving real-world problems. The internship also emphasized continuous learning, critical thinking, and the importance of adapting to rapidly evolving technological advancements.

This report presents the work carried out during the BSERC–ISRO Def-Space Summer Internship Programme. It documents the objectives of the internship, the activities performed, the tools utilized, the assignments completed, the knowledge gained, and the overall learning experience throughout the programme. The report reflects the practical understanding acquired during the internship while highlighting the importance of AI-assisted learning in enhancing technical competence and professional development.

1.1 About BSERC

The **Bharat Space Education Research Centre (BSERC)** is an educational and research-oriented organization dedicated to promoting scientific learning, innovation, and technical excellence among students across India. BSERC conducts various educational initiatives, workshops, certification programmes, research activities, and internship programmes in collaboration with academic institutions and industry experts to encourage the practical application of science and technology. The organization aims to create opportunities for students to develop technical skills in emerging domains such as Artificial Intelligence, Machine Learning, Data Science, Robotics, Internet of Things (IoT), Space Science, Defence Technologies, and other advanced engineering disciplines. Through structured learning programmes and project-based activities, BSERC enables students to strengthen their academic knowledge while preparing them for future research and industrial careers.

1.2 About ISRO

The **Indian Space Research Organisation (ISRO)** is India's national space agency and one of the world's leading organizations in space research, satellite technology, launch vehicle development, remote sensing, navigation systems, planetary exploration, and scientific missions. Established in 1969, ISRO has significantly contributed to India's technological advancement by developing cost-effective space missions and innovative satellite applications that support communication, weather

forecasting, disaster management, agriculture, education, healthcare, navigation, and national development.

Apart from its achievements in space exploration, ISRO actively promotes scientific education, research, innovation, and skill development among students through various outreach programmes, internships, educational initiatives, and collaborative learning opportunities. These programmes encourage young engineers and researchers to contribute towards the future development of India's space and technological capabilities.

1.3 About the Def-Space Summer Internship Programme

The **BSERC–ISRO Def-Space Summer Internship Programme 2026** is an industry-oriented technical internship designed to provide students with practical exposure to modern technologies through structured learning modules, assignments, and AI-assisted educational activities. The internship integrates theoretical concepts with practical applications, enabling students to understand the role of Artificial Intelligence and emerging technologies in solving real-world engineering challenges.

Throughout the internship, participants engaged in multiple technical learning activities that encouraged analytical thinking, independent research, documentation, and professional communication. The internship emphasized the effective use of AI-assisted tools such as NotebookLM for learning, knowledge organization, and technical content analysis while promoting ethical and responsible usage of Artificial Intelligence.

The programme also provided valuable insights into current industry expectations, helping students identify technical competencies, understand career pathways, and develop structured learning plans for continuous professional growth.

1.4 Internship Objectives

The primary objectives of the BSERC–ISRO Def-Space Summer Internship Programme are:

- To provide practical exposure to Artificial Intelligence and emerging technologies.
- To bridge the gap between academic learning and industry expectations.
- To strengthen technical, analytical, and problem-solving skills.
- To introduce students to AI-assisted learning methodologies.
- To improve professional communication and technical documentation abilities.
- To develop awareness of current technological trends in AI, Machine Learning, and Defence applications.
- To encourage independent learning and continuous skill development.
- To prepare students for future academic research and industrial careers.

1.5 Scope of the Internship

The internship focuses on developing technical competence through structured learning activities, assignments, and AI-assisted educational tools. It covers concepts related to Artificial Intelligence, Machine Learning, Large Language Models, career readiness, technical communication, and modern learning methodologies. The programme enables participants to understand practical applications of AI while enhancing professional skills required for future engineering and research careers.

The scope of this report is limited to the activities completed during the internship, including the assignments performed using NotebookLM, the analysis of AI/ML career requirements, and the educational study of Large Language Models. The report documents the methodologies followed, observations made, learning outcomes achieved, and the overall experience gained throughout the internship.

1.6 Expected Learning Outcomes

Upon successful completion of the internship, the following learning outcomes were achieved:

- Improved understanding of Artificial Intelligence and Machine Learning concepts.
- Practical experience in using AI-assisted learning platforms such as NotebookLM.
- Enhanced ability to analyze industry job requirements and identify skill gaps.
- Better understanding of Large Language Models and their practical applications.
- Improved technical writing, documentation, and communication skills.
- Strengthened analytical thinking and problem-solving abilities.
- Increased awareness of emerging technologies and future career opportunities.
- Development of a structured learning approach for continuous professional growth.

CHAPTER 2

INTERNSHIP ACTIVITIES

2.1 Overview of the Internship

The **BSERC–ISRO Def-Space Summer Internship Programme 2026** was designed to provide participants with practical exposure to emerging technologies while strengthening their technical and professional competencies. Throughout the internship, various learning activities, technical sessions, and assignments were conducted to encourage participants to explore Artificial Intelligence, Machine Learning, Large Language Models, and modern digital tools used in engineering and research. Unlike conventional classroom learning, the internship emphasized self-learning, analytical thinking, and the effective utilization of Artificial Intelligence-assisted educational platforms. Participants were encouraged to analyze technical information, understand industry expectations, and communicate complex concepts in a simplified and structured manner. This approach helped bridge the gap between theoretical knowledge and practical application while promoting independent learning.

The internship also focused on improving documentation skills, technical writing, and career preparedness by assigning activities that required critical thinking, research, and structured presentation of technical information. Each assignment was designed to simulate real-world scenarios where participants were expected to analyze information, draw meaningful conclusions, and present their findings professionally.

Overall, the internship provided a comprehensive learning environment that enabled participants to strengthen their technical foundation while developing essential professional skills required in today's rapidly evolving technology landscape.

2.2 Internship Workflow

The internship activities were carried out in a structured sequence to facilitate progressive learning. The overall workflow followed during the internship is illustrated below.

Internship Registration



Orientation Session



Access to Learning Resources



Technical Learning Modules



Assignment 1 – AI/ML Engineer Job Ready Decoder



Assignment 2 – Large Language Models using NotebookLM



Analysis and Documentation



Final Report Preparation



Submission

(Insert a flowchart illustrating the above workflow if required.)

2.3 Tools and Technologies Used

Several digital platforms and AI-assisted tools were utilized throughout the internship to support learning, research, documentation, and analysis. Each tool served a specific purpose in completing the assigned activities efficiently.

Google NotebookLM

Google NotebookLM was the primary AI-assisted learning platform used during the internship. It enabled document analysis, content summarization, concept explanation, and information organization based on uploaded reference materials. The platform played a significant role in both internship assignments by assisting with resume analysis, job description comparison, and educational explanations of technical concepts.

PDF Documents

Various PDF documents, including technical notes, study materials, and industry job descriptions, were used as reference sources for analysis and learning activities.

Microsoft Word

Microsoft Word was used to prepare technical documentation, organize assignment responses, and develop the final internship report in a structured and professional format.

Internet Resources

Authentic online resources were referred to for understanding emerging technologies, validating technical concepts, and gaining additional knowledge related to Artificial Intelligence and Machine Learning.

2.4 Internship Assignments

The internship consisted of two major technical assignments that emphasized the practical use of Artificial Intelligence tools for learning and professional development.

Assignment 1 – AI/ML Engineer Job Ready Decoder

The first assignment focused on understanding the expectations of an AI/ML Engineer role. Multiple industry job descriptions were analyzed using NotebookLM and compared with the participant's resume. The activity helped identify existing technical strengths, missing competencies, skill gaps, interview preparation areas, and future learning requirements.

The assignment also required preparing interview responses and developing a structured learning roadmap to enhance employability and career readiness.

Assignment 2 – Large Language Models using NotebookLM

The second assignment focused on studying the fundamentals of Large Language Models (LLMs), particularly Llama 3. Participants used NotebookLM to understand technical concepts and explain them for audiences with different levels of technical knowledge.

The activity demonstrated how Artificial Intelligence can assist in learning while also highlighting the importance of human reasoning, creativity, and effective communication when interpreting AI-generated content.

2.5 Skills Developed During the Internship

The internship contributed significantly to the development of both technical and professional competencies.

Technical Skills

- Artificial Intelligence Fundamentals

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- Machine Learning Concepts
 - Large Language Models (LLMs)
 - Google NotebookLM
 - AI-assisted Research
 - Resume Analysis
 - Job Description Analysis
 - Technical Documentation
 - Information Organization

Professional Skills

- Technical Communication
- Analytical Thinking
- Critical Thinking
- Problem Solving
- Research Skills
- Documentation
- Professional Writing
- Time Management
- Continuous Learning

2.6 Key Learning Experience

One of the most valuable aspects of the internship was understanding how Artificial Intelligence can support learning without replacing human reasoning. The internship demonstrated that AI tools are most effective when used responsibly to analyze information, generate insights, and improve productivity while allowing learners to critically evaluate and refine the generated content.

The assignments also provided practical exposure to career preparation by encouraging participants to assess their technical competencies against current industry expectations. Through structured

analysis and reflection, participants gained a clearer understanding of the skills required for professional growth in Artificial Intelligence and Machine Learning.

Additionally, the internship strengthened documentation, communication, and presentation skills, which are essential for both academic research and industrial environments. The overall experience promoted independent learning, adaptability, and lifelong skill development.

2.7 Chapter Summary

This chapter presented an overview of the activities carried out during the BSERC–ISRO Def-Space Summer Internship Programme. It described the internship workflow, tools and technologies used, assignments completed, skills developed, and the overall learning experience. These activities established the foundation for the detailed discussion of the two internship assignments presented in the following chapters.

CHAPTER 3

ASSIGNMENT 1 – AI/ML ENGINEER JOB READY DECODER

3.1 Introduction

One of the primary assignments of the BSERC–ISRO Def-Space Summer Internship Programme was the **AI/ML Engineer Job Ready Decoder**. The objective of this assignment was to evaluate an individual's readiness for an Artificial Intelligence and Machine Learning Engineer role by comparing real industry job descriptions with the participant's current resume.

Instead of manually analyzing multiple job descriptions, **Google NotebookLM** was used as an AI-assisted document analysis tool. Three authentic AI/ML Engineer job descriptions and the participant's resume were uploaded into NotebookLM. The platform analyzed all the documents collectively and generated insights regarding technical skills, programming languages, soft skills, qualifications, potential interview questions, and future learning recommendations.

This assignment demonstrated how Generative AI can be utilized for career planning, resume evaluation, interview preparation, and skill-gap analysis.

3.2 Objective of the Assignment

The objectives of this assignment were:

- To analyze multiple AI/ML Engineer job descriptions.
- To compare industry requirements with the participant's resume.
- To identify strengths and missing skills.

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- To evaluate overall career readiness.
 - To generate technical interview questions.
 - To prepare personalized interview responses.
 - To develop a structured learning roadmap for future improvement.

3.3 Methodology

The assignment was completed using the following workflow:

1. Selected **AI/ML Engineer** as the target career role.
2. Collected three authentic AI/ML Engineer job descriptions from different organizations.
3. Uploaded all job descriptions along with the participant's resume into Google NotebookLM.
4. Used NotebookLM prompts to compare the documents.
5. Generated:
 - Skill ranking
 - Skill gap analysis
 - Technical observations
 - Interview questions
 - Suggested answers
 - 30-day learning plan
6. Analyzed the generated output and documented the findings.

3.4 Resources Used

Resource	Purpose
AI/ML Engineer Job Description 1	Industry requirement analysis
AI/ML Engineer Job Description 2	Industry requirement analysis
AI/ML Engineer Job Description 3	Industry requirement analysis
Personal Resume	Candidate profile
Google NotebookLM	AI-assisted document comparison

3.5 Skill Ranking Generated by NotebookLM

After analyzing all three job descriptions together with the uploaded resume, NotebookLM generated the following ranked list of skills based on how frequently they appeared across the job descriptions.

Rank	Category	Skill / Qualification	Mention Frequency	Status in Resume
1	Technical	Machine Learning (ML)	3 of 3 JDs	Present
2	Technical	Artificial Intelligence (AI) Development	3 of 3 JDs	Present
3	Technical	Data Science / Data Analysis	3 of 3 JDs	Present
4	Soft Skill	Teamwork / Collaboration	3 of 3 JDs	Present
5	Programming Language	Python Programming	2 of 3 JDs	Present
6	Technical	Generative AI (GenAI)	2 of 3 JDs	Present
7	Technical	Deployment / Production Pipelines	2 of 3 JDs	Present

8	Technical	Feature Engineering / Data Exploration	2 of 3 JDs	Present
9	Technical	Chatbots / Conversational AI	2 of 3 JDs	Missing
10	Soft Skill	Analytical / Problem Solving	2 of 3 JDs	Present
11	Qualification	Bachelor's Degree (Computer Science)	2 of 3 JDs	Present
12	Technical	Mathematics & Statistics	2 of 3 JDs	Present
13	Technical	Testing & Troubleshooting	2 of 3 JDs	Present

3.6 Analysis of the Generated Results

The analysis performed by NotebookLM revealed that the participant's profile demonstrates strong alignment with the core requirements expected from an AI/ML Engineer.

Top Tier Alignment (3 out of 3 Job Descriptions)

The highest-ranked competencies identified across all three job descriptions were:

- Machine Learning
- Artificial Intelligence Development
- Data Science
- Teamwork and Collaboration

These skills were also clearly reflected in the participant's resume through technical projects, research experience, and collaborative activities, indicating a strong match between the current profile and industry expectations.

Technical Strengths

NotebookLM highlighted several additional strengths present in the participant's resume:

- Experience with Generative AI technologies, particularly Llama 3.
- Strong proficiency in Python programming.
- Familiarity with Docker-based deployment and containerized environments.

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- Practical exposure to production-oriented AI development.

These strengths suggest that the participant possesses a solid technical foundation suitable for AI/ML Engineer roles.

Key Missing Skill

The primary skill identified as missing was:

Chatbots / Conversational AI

Although the participant has worked extensively with Llama 3 and Generative AI, the resume did not explicitly include chatbot or virtual assistant development projects.

Qualification Analysis

NotebookLM also verified that the participant satisfies the educational qualification expected by most organizations through an ongoing Bachelor's degree in Computer Science.

However, the analysis recommended obtaining additional professional AI certifications to further strengthen the profile.

3.7 Predicted Interview Questions

Based on the uploaded resume and the three job descriptions, NotebookLM generated the following interview questions that are likely to be asked during an AI/ML Engineer interview.

Technical Questions

Question 1

How did you integrate Llama 3 into the UniShield framework, and what advantages did it provide over traditional threat intelligence methods?

Question 2

Why did you choose the Double Deep Q-Network (DDQN) architecture for adaptive deception, and how did you design the reward function?

Question 3

When developing cybersecurity detection models, how did you perform feature engineering to distinguish between benign traffic and supply-chain attacks?

Question 4

How would you deploy AI models in production using Docker and container technologies?

Question 5

How can Software Defined Networking (SDN) improve large-scale AI pipelines?

Behavioral Questions

Question 6

Describe a situation where you demonstrated leadership or teamwork during a technical project.

Question 7

Tell us about a time when an AI model failed to achieve the expected performance. How did you troubleshoot and improve it?

Question 8

How do you stay updated with the latest developments in Artificial Intelligence and Machine Learning?

3.8 Selected Interview Questions and Answers

Among the generated questions, the following three were selected because they best matched the participant's experience.

Question 1

How did you integrate Llama 3 into UniShield, and what advantages did it provide?

Answer

During my research internship at the Centre for Computer Networks and Cyber Security (CCNCS), I worked on the UniShield framework, where Llama 3 was integrated into the adaptive deception layer. Instead of relying only on predefined rules, Llama 3 analyzed attacker behavior and generated context-aware responses for different attack scenarios. This improved threat analysis, supported intelligent interaction with attackers, and helped collect more meaningful threat intelligence while adapting to evolving cyber threats.

Confidence Level: High

Question 2

Why did you choose DDQN for adaptive deception?

Answer

We selected the Double Deep Q-Network because it reduces the overestimation problem found in traditional Q-learning and provides more stable learning. In UniShield, the DDQN agent learned to choose the most effective deception strategy based on the attacker's behavior. The reward function encouraged successful attacker engagement, high-quality threat intelligence collection, and efficient resource utilization while penalizing ineffective or unnecessary actions.

Confidence Level: High

Question 3

Tell us about a time when an AI model failed to meet performance expectations.

Answer

During the development of UniShield, some detection models initially showed lower accuracy because of class imbalance and inconsistent feature distributions in the datasets. I analyzed the preprocessing pipeline, verified the feature engineering process, balanced the dataset, and tuned the model parameters. After retraining and validating the models, the performance improved significantly, resulting in better detection accuracy and more reliable predictions.

Confidence Level: Medium

3.9 Thirty-Day Learning Plan

Based on the identified skill gaps, NotebookLM generated the following learning roadmap.

Table 3.2 Thirty-Day Learning Plan

Week	Learning Goal	Planned Activity
Week 1	Conversational AI	Complete a course on chatbots using LangChain, Hugging Face, or OpenAI APIs and build a simple chatbot.
Week 2	MLflow & MLOps	Learn experiment tracking, model versioning, and deployment using MLflow.
Week 3	Production Deployment	Containerize an AI model using Docker and deploy it locally using FastAPI or Flask.
Week 4	Portfolio Enhancement	Integrate the chatbot and deployment project into GitHub, update the resume, and practice AI/ML interview questions.

3.10 Learning Outcomes

The AI/ML Engineer Job Ready Decoder assignment provided valuable insights into the expectations of the current AI/ML industry. By leveraging Google NotebookLM for document analysis, the participant gained a structured understanding of industry-demanded skills, resume alignment, interview preparation, and future learning priorities. The assignment also demonstrated the practical application of Generative AI as a tool for career planning, self-assessment, and professional development. The generated learning roadmap offers a clear direction for enhancing technical competencies and improving readiness for future AI/ML engineering roles.

CHAPTER 4

ASSIGNMENT 2 – UNDERSTANDING LARGE LANGUAGE MODELS USING NOTEBOOKLM

4.1 Introduction

The second assignment of the **BSERC–ISRO Def-Space Summer Internship Programme** focused on understanding **Large Language Models (LLMs)** and exploring how Artificial Intelligence can explain the same concept to audiences with different levels of technical knowledge. The assignment was designed to demonstrate the capabilities of **Google NotebookLM** in analyzing educational material and generating explanations tailored to the background of the reader.

Unlike traditional learning methods where a single explanation is provided for all learners, modern AI systems can dynamically adapt their explanations according to the user's knowledge level. This assignment emphasized the importance of effective technical communication and highlighted the role of AI-assisted educational tools in simplifying complex concepts without losing the core meaning. The activity also encouraged participants to critically evaluate AI-generated responses by identifying missing details, recognizing oversimplifications, and developing an improved explanation based on their own understanding.

4.2 Objective of the Assignment

The primary objectives of this assignment were:

- To understand the fundamentals of Large Language Models (LLMs).

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- To explore the capabilities of Google NotebookLM for AI-assisted learning.
 - To analyze how AI adapts explanations for different audiences.
 - To identify the strengths and limitations of AI-generated educational content.
 - To improve AI-generated explanations through human reasoning and creativity.

4.3 Assignment Methodology

The assignment was carried out using the following procedure:

1. A study document containing information about **Large Language Models (LLMs)** was uploaded to **Google NotebookLM**.
2. NotebookLM analyzed the uploaded content and generated explanations suitable for different audiences.
3. Three explanations were requested for:
 - A 12th-grade student
 - A working professional
 - A five-year-old child
4. The generated explanations were compared to identify differences in technical depth and presentation.
5. The participant analyzed what information disappeared, what became oversimplified, and how the explanation could be further improved.
6. Finally, an original one-sentence explanation was created to communicate the concept in a simple and engaging manner.

4.4 Explanation of Large Language Models for Different Audiences

4.4.1 Explanation for a 12th-Grade Student

NotebookLM explained that **Large Language Models (LLMs)** represent a significant advancement in Artificial Intelligence compared to traditional rule-based systems. Unlike older AI models

designed for specific tasks, LLMs are **foundation models** capable of performing multiple language-related tasks such as summarizing text, translating languages, answering questions, and generating computer code.

The explanation introduced technical concepts including **deep neural networks**, **Transformer architecture**, and **next-word prediction**. It also described how these models are trained on billions of words collected from books, articles, and internet resources, enabling them to understand context and generate human-like responses.

4.4.2 Explanation for a Working Professional

For a working professional, NotebookLM emphasized the practical applications of LLMs rather than their internal architecture.

The explanation highlighted that Large Language Models can automate business processes, improve productivity, generate content, support intelligent virtual assistants, and assist in domains such as cybersecurity analysis.

NotebookLM also introduced important practical considerations including:

- Hallucinations
- Data bias
- Human verification of AI-generated responses

This explanation focused on how organizations can benefit from LLMs while understanding their limitations.

4.4.3 Explanation for a Five-Year-Old Child

For a young child, NotebookLM simplified the concept completely by describing an LLM as:

"A super-smart computer that has read almost every book in the world so it can talk to you, answer your questions, and tell you stories just like a person would."

The explanation avoided all technical terminology and instead used a simple analogy that could be easily understood by children.

4.5 Comparative Analysis of the Generated Explanations

One of the key learning outcomes of this assignment was observing how Artificial Intelligence automatically adjusts explanations based on the intended audience.

Table 4.1 Comparison of the Three Explanations

Audience	Characteristics of the Explanation
12th-Grade Student	Included technical concepts such as deep neural networks, Transformer architecture, foundation models, and next-word prediction while maintaining understandable language.
Working Professional	Focused on business applications, productivity, intelligent assistants, cybersecurity, hallucinations, and practical deployment rather than detailed technical theory.
Five-Year-Old Child	Used a simple analogy of a "super-smart computer" while completely removing technical terminology and complex concepts.

The comparison clearly demonstrated the flexibility of AI-assisted educational systems in adapting content according to the learner's background and level of understanding.

4.6 Information Removed for Less Technical Audiences

As the target audience became less technical, NotebookLM progressively removed several advanced concepts from the explanation.

The omitted concepts included:

- Transformer Architecture
- Deep Neural Networks
- Token Prediction Process

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- Training on Billions of Words
 - Foundation Model Concept
 - Technical Inference Mechanisms
 - AI Reasoning Process
 - Ethical Challenges
 - Research Applications

The removal of these concepts improved readability but reduced the technical depth of the explanation.

4.7 Oversimplified Concepts

Although the simplified explanations were easy to understand, they also introduced several oversimplifications.

NotebookLM's explanation implied that:

- The model had "read almost every book in the world."
- The system understands language exactly like humans.
- The explanation ignored hallucinations.
- AI bias was not discussed.
- Human verification requirements were omitted.
- The learning process of the model was not explained.

These observations demonstrate that AI-generated educational content should always be critically evaluated before being used for teaching or decision-making.

4.8 Human-Improved Explanation (Beat the AI)

As the final activity of the assignment, participants were asked to improve the AI-generated explanation using their own creativity and understanding.

The following explanation was developed:

"A Large Language Model is like a very smart robot friend that learns from lots of books and conversations so it can answer your questions, tell stories, and help you solve problems."

This explanation preserves the simplicity required for beginners while conveying the primary purpose of a Large Language Model in a more engaging and relatable manner.

4.9 Learning Outcomes

This assignment demonstrated that Artificial Intelligence can effectively personalize educational content according to the learner's level of expertise. By using Google NotebookLM, it became evident that AI can simplify complex concepts, organize information efficiently, and support self-directed learning. At the same time, the exercise highlighted the importance of human judgment in verifying AI-generated content, recognizing oversimplifications, and improving explanations where necessary.

The assignment also strengthened skills in technical communication, critical thinking, content evaluation, and AI-assisted learning. It reinforced the understanding that while AI is a powerful educational tool, meaningful learning still depends on human reasoning, domain knowledge, and the ability to critically assess generated information before applying it in academic or professional contexts.

CHAPTER 5

OVERALL LEARNING EXPERIENCE

5.1 Introduction

The BSERC–ISRO Def-Space Summer Internship Programme provided an enriching learning experience by exposing participants to Artificial Intelligence, Machine Learning, Large Language Models, and AI-assisted educational technologies. Throughout the internship, various structured assignments and learning activities encouraged the practical application of technical knowledge while strengthening analytical thinking, communication skills, and professional development. Unlike conventional academic exercises, the internship emphasized independent learning, critical analysis, and the responsible use of Artificial Intelligence tools for solving real-world problems. The assignments not only improved technical understanding but also highlighted the importance of continuous learning in rapidly evolving technological domains.

5.2 Technical Knowledge Gained

The internship significantly enhanced my understanding of several important concepts in Artificial Intelligence and related technologies.

The major technical knowledge acquired during the internship includes:

- Fundamentals of Artificial Intelligence and Machine Learning.
- Understanding the architecture and applications of Large Language Models.
- Practical use of Google NotebookLM for document analysis and AI-assisted learning.
- Resume analysis using AI-powered tools.

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- Skill gap analysis based on real-world industry job descriptions.
 - AI-assisted interview preparation.
 - Technical documentation and report writing.
 - Effective interpretation of AI-generated content.

The internship also demonstrated how Artificial Intelligence can simplify complex learning processes while supporting research, knowledge organization, and decision-making.

5.3 Professional Skills Developed

In addition to technical knowledge, the internship contributed significantly to the development of professional competencies that are essential for future academic and industrial careers.

The professional skills developed include:

- Technical communication
- Analytical thinking
- Critical evaluation of AI-generated information
- Problem-solving
- Documentation and report writing
- Independent learning
- Information organization
- Time management
- Adaptability to new technologies
- Professional presentation of technical information

These skills are valuable not only in Artificial Intelligence but also across multidisciplinary engineering domains.

5.4 Challenges Encountered During the Internship

Several challenges were encountered while completing the internship assignments.

One of the primary challenges was understanding how to effectively utilize AI-assisted tools without relying entirely on AI-generated responses. Although NotebookLM produced comprehensive explanations and analyses, it was necessary to carefully verify, interpret, and refine the generated content to ensure accuracy and clarity.

Another challenge involved understanding the expectations of the AI/ML industry by comparing multiple job descriptions simultaneously. Initially, identifying common skill requirements manually would have required considerable effort; however, the use of NotebookLM simplified this process while still requiring careful interpretation of the generated results.

Explaining technical concepts for audiences with different educational backgrounds also presented an interesting challenge. The assignment highlighted the importance of balancing technical accuracy with simplicity and demonstrated that effective communication requires adapting explanations according to the audience's level of understanding.

5.5 How the Challenges Were Addressed

The challenges encountered during the internship were addressed through a combination of structured learning, careful analysis, and continuous verification of AI-generated content.

Technical concepts that were initially difficult to understand were studied using multiple learning resources before accepting the AI-generated explanations. Every response produced by NotebookLM was critically reviewed and refined to improve clarity and correctness.

For the job readiness assignment, multiple job descriptions were carefully compared to understand recurring industry expectations, enabling the preparation of a realistic learning roadmap. Similarly, for the Large Language Model assignment, the AI-generated explanations were evaluated from different perspectives before producing an improved explanation using personal understanding.

This process emphasized that Artificial Intelligence should be viewed as an intelligent learning assistant rather than a replacement for human reasoning.

5.6 Key Outcomes of the Internship

The internship resulted in several significant academic and professional outcomes.

The major outcomes include:

- Improved understanding of Artificial Intelligence and Machine Learning concepts.
- Practical experience using Google NotebookLM for AI-assisted learning.
- Enhanced awareness of AI/ML industry expectations.
- Better understanding of Large Language Models and their applications.
- Improved resume evaluation and interview preparation skills.
- Development of structured learning and self-assessment strategies.
- Increased confidence in using modern AI tools responsibly.
- Strengthened documentation and technical writing abilities.

These outcomes collectively contributed to both technical competence and professional growth.

5.7 Overall Reflection

The BSERC–ISRO Def-Space Summer Internship Programme provided an excellent platform to explore the practical applications of Artificial Intelligence in education and professional development. The internship demonstrated that AI tools such as NotebookLM can significantly improve productivity, organize information efficiently, and simplify complex technical concepts when used appropriately.

At the same time, the internship reinforced the importance of human judgment in validating AI-generated content. Critical thinking, domain knowledge, and effective communication remain essential for producing accurate, meaningful, and reliable technical work.

Overall, the internship strengthened my confidence in applying Artificial Intelligence as a learning companion while encouraging continuous learning, innovation, and professional development.

5.8 Chapter Summary

This chapter summarized the overall learning experience gained during the BSERC–ISRO Def-Space Summer Internship Programme. It discussed the technical knowledge acquired, professional skills developed, challenges encountered, solutions adopted, and the overall impact of the internship on personal and professional growth. The internship successfully bridged theoretical concepts with practical applications and provided valuable exposure to AI-assisted learning methodologies, preparing participants for future academic, research, and industry opportunities.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The **BSERC–ISRO Def-Space Summer Internship Programme 2026** provided an excellent opportunity to strengthen both technical knowledge and professional skills through structured learning activities and AI-assisted educational assignments. Throughout the internship, I gained practical exposure to emerging technologies, particularly in the fields of **Artificial Intelligence (AI)**, **Machine Learning (ML)**, and **Large Language Models (LLMs)**, while understanding their growing significance in solving real-world problems.

The internship assignments encouraged the effective use of **Google NotebookLM** as an AI-powered learning platform for document analysis, information organization, technical explanation, and career preparation. The **AI/ML Engineer Job Ready Decoder** assignment provided valuable insights into current industry expectations by comparing multiple job descriptions with my resume. This exercise helped identify my technical strengths, recognize skill gaps, understand interview expectations, and prepare a structured learning roadmap for future career development.

The second assignment focused on understanding **Large Language Models** and demonstrated how AI can adapt explanations according to different audiences. By comparing AI-generated explanations and critically evaluating their strengths and limitations, I developed a better understanding of the importance of technical communication, human reasoning, and responsible use of Artificial Intelligence.

Overall, the internship successfully enhanced my analytical thinking, technical documentation skills, communication abilities, and confidence in using AI-assisted learning tools. More importantly, it reinforced the idea that Artificial Intelligence is most effective when used as a collaborative learning assistant supported by human judgment, creativity, and critical thinking. The knowledge and experience gained through this internship will serve as a strong foundation for my future academic, research, and professional journey.

6.2 Future Scope

The knowledge and skills acquired during the BSERC–ISRO Def-Space Summer Internship provide a strong foundation for continuous learning and future career development. Based on the outcomes of the internship, the following areas have been identified for future improvement and exploration:

- Strengthen practical expertise in **Machine Learning Operations (MLOps)** and AI model deployment.
- Develop hands-on projects involving **Conversational AI**, intelligent virtual assistants, and chatbot applications.
- Explore advanced **Generative AI** frameworks and emerging Large Language Models for real-world applications.
- Gain industry-recognized certifications in Artificial Intelligence, Machine Learning, and Cloud Technologies.
- Build and deploy end-to-end AI solutions using modern tools such as Docker, FastAPI, and cloud platforms.
- Continue improving technical communication, documentation, and presentation skills for professional and research environments.
- Expand knowledge of ethical AI, responsible AI practices, and trustworthy AI systems.
- Participate in additional internships, workshops, hackathons, and research activities to strengthen practical exposure and industry readiness.

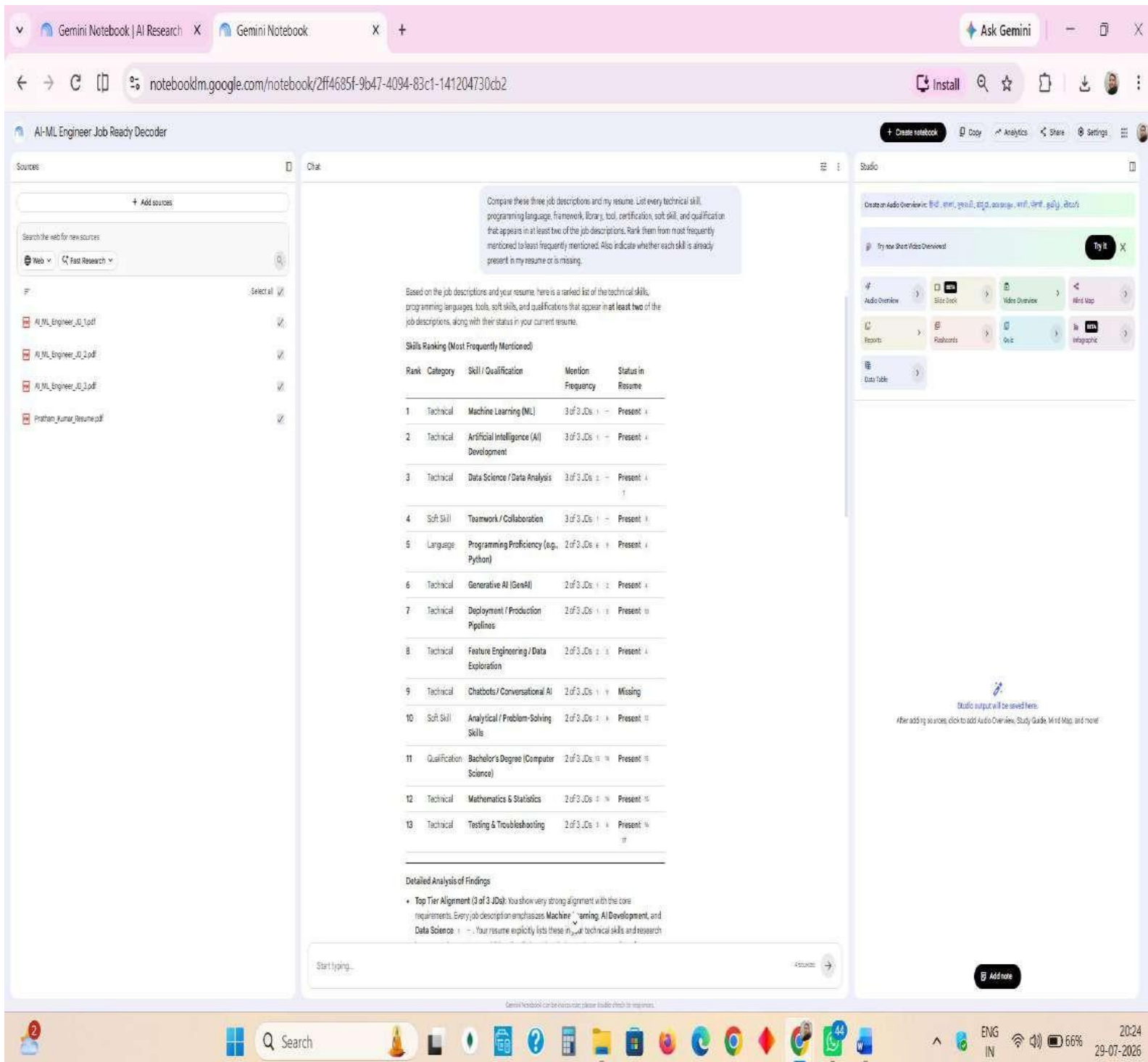
The internship has inspired a commitment to lifelong learning and continuous skill enhancement. By building upon the knowledge gained during this programme, I aim to contribute meaningfully to the advancement of Artificial Intelligence and emerging technologies while preparing for future opportunities in research, innovation, and the technology industry.

REFERENCES

The following resources were referred to during the completion of the internship assignments and preparation of this report.

1. Google. **NotebookLM – AI-Powered Research and Note-Taking Assistant.**
<https://notebooklm.google/>
2. Meta AI. **Llama 3: Open Foundation and Instruction Models.** <https://ai.meta.com/llama/>
3. Google AI. **Introduction to Large Language Models.**
<https://developers.google.com/machine-learning/resources>
4. Accenture Careers. **AI/ML Engineer Job Description.**
5. ArchLynk Careers. **Artificial Intelligence / Machine Learning Engineer Job Description.**
6. Indeed Career Guide. **AI Engineer Job Description and Career Guide.**
7. Bharat Space Education Research Centre (BSERC). **Def-Space Summer Internship Programme 2026.**
8. Indian Space Research Organisation (ISRO). **Official Website.** <https://www.isro.gov.in/>
9. Course materials, study notes, and assignment guidelines provided during the **BSERC–ISRO Def-Space Summer Internship Programme 2026.**
10. Personal resume and assignment outputs generated using **Google NotebookLM** during the internship activities.

SCREENSHOTS



The screenshot displays the Gemini NotebookLM interface with a notebook titled "AI-ML Engineer Job Ready Decoder". The interface is divided into three main sections: Sources, Chat, and Studio.

Sources: Lists the documents used for analysis: "AI_ML_Engineer_JD_1.pdf", "AI_ML_Engineer_JD_2.pdf", "AI_ML_Engineer_JD_3.pdf", and "Prathan_Jumar_Resume.pdf".

Chat: Contains a user prompt: "Compare these three job descriptions and my resume. List every technical skill, programming language, framework, library, tool, certification, soft skill, and qualification that appears in at least two of the job descriptions. Rank them from most frequently mentioned to least frequently mentioned. Also indicate whether each skill is already present in my resume or is missing." The response includes a summary and a table of skills.

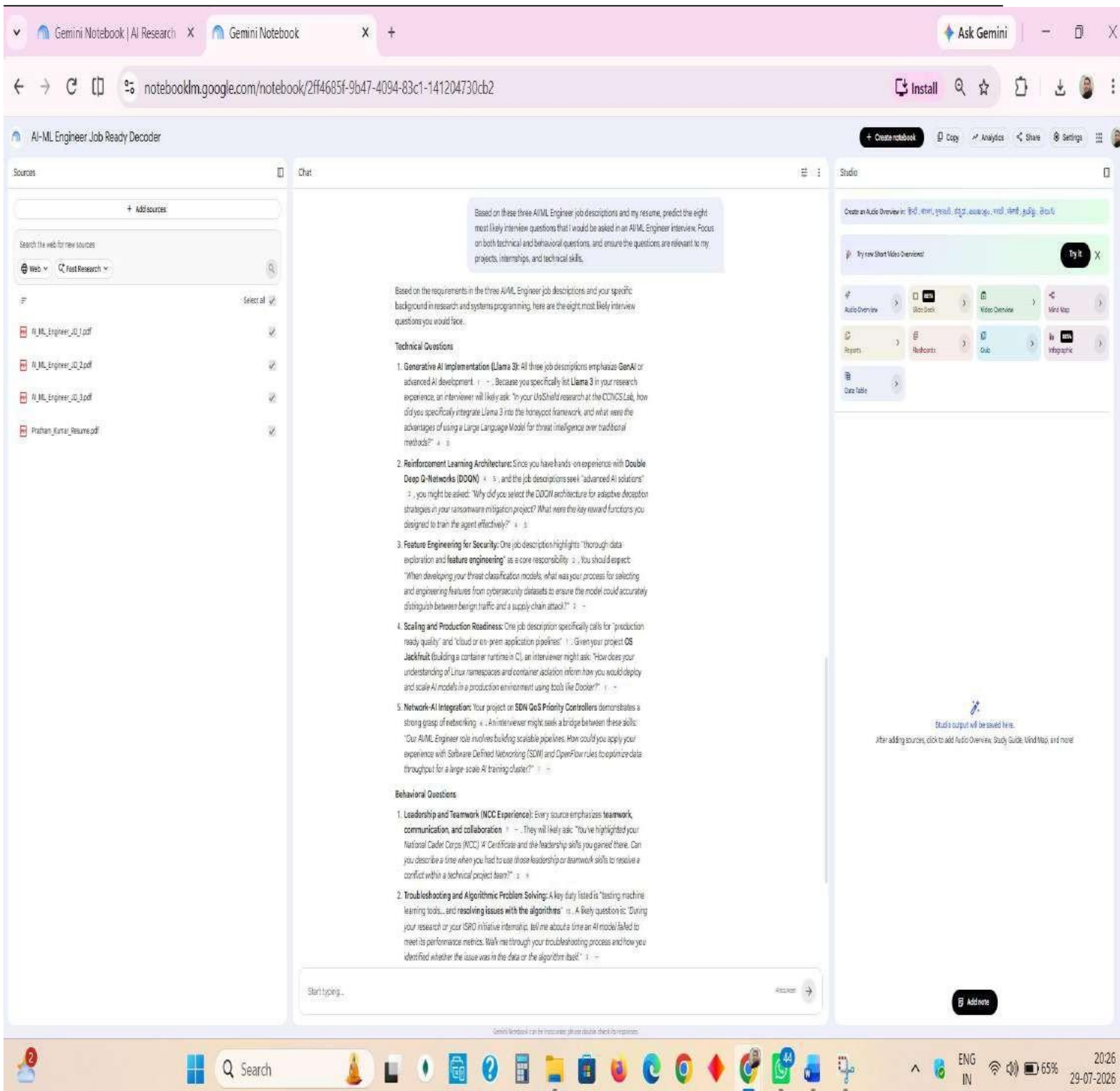
Skills Ranking (Most Frequently Mentioned):

Rank	Category	Skill / Qualification	Mention Frequency	Status in Resume
1	Technical	Machine Learning (ML)	3 of 3 JDs	Present
2	Technical	Artificial Intelligence (AI) Development	3 of 3 JDs	Present
3	Technical	Data Science / Data Analysis	3 of 3 JDs	Present
4	Soft Skill	Teamwork / Collaboration	3 of 3 JDs	Present
5	Language	Programming Proficiency (e.g., Python)	2 of 3 JDs	Present
6	Technical	Generative AI (GenAI)	2 of 3 JDs	Present
7	Technical	Deployment / Production Pipelines	2 of 3 JDs	Present
8	Technical	Feature Engineering / Data Exploration	2 of 3 JDs	Present
9	Technical	Chatbots / Conversational AI	2 of 3 JDs	Missing
10	Soft Skill	Analytical / Problem-Solving Skills	2 of 3 JDs	Present
11	Qualification	Bachelor's Degree (Computer Science)	2 of 3 JDs	Present
12	Technical	Mathematics & Statistics	2 of 3 JDs	Present
13	Technical	Testing & Troubleshooting	2 of 3 JDs	Present

Detailed Analysis of Findings:

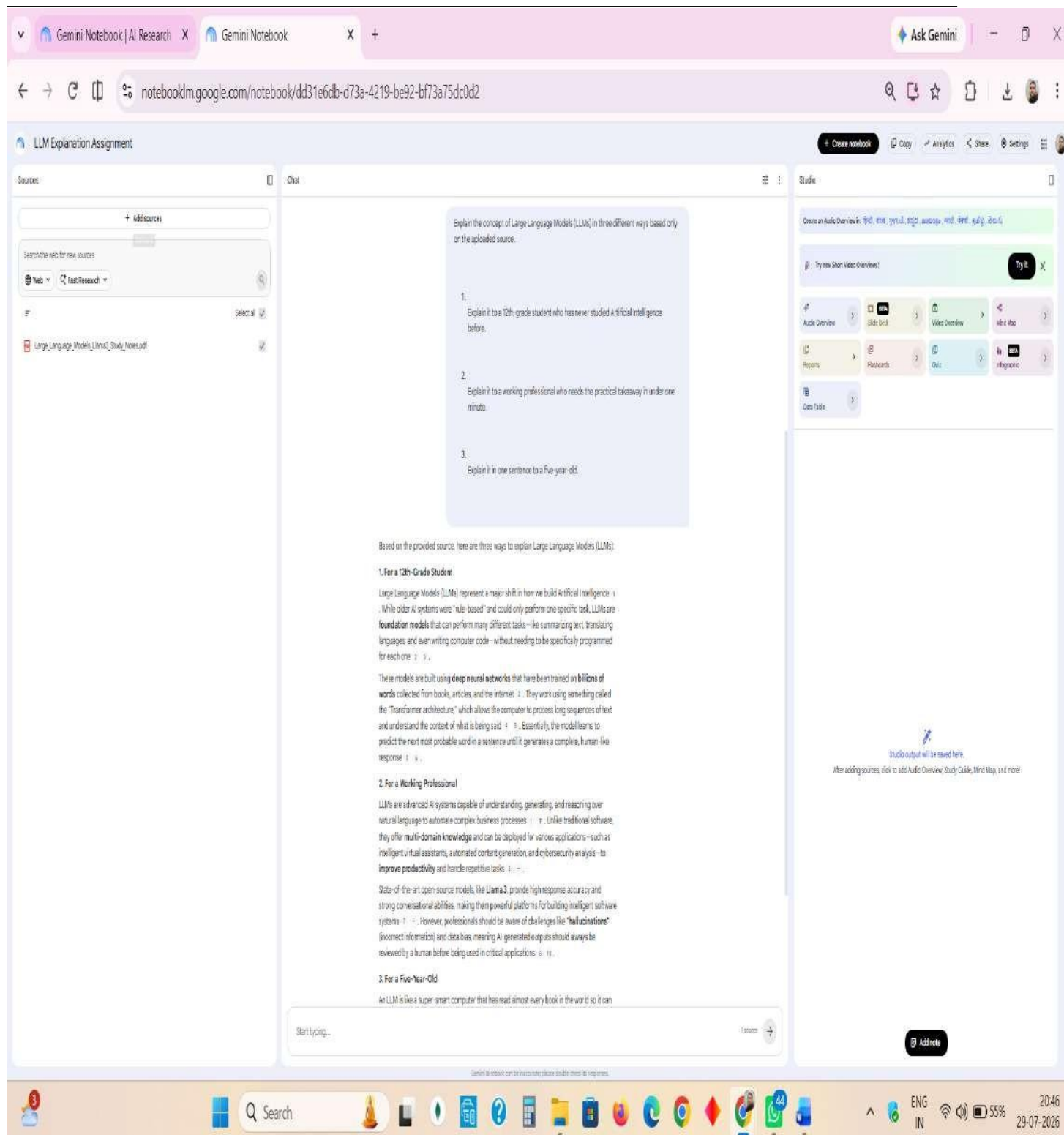
- Top Tier Alignment (3 of 3 JDs):** You show very strong alignment with the core requirements. Every job description emphasizes Machine Learning, AI Development, and Data Science. Your resume explicitly lists these in your technical skills and research.

Studio: Offers options to create an audio overview, try a short video overview, or add more content like reports, rubrics, or data table.



The screenshot displays the Gemini NotebookLM web application. The interface is divided into three main sections: Sources, Chat, and Studio.

- Sources:** On the left, there is a list of sources used for the chat. It includes four PDF files named "AI_ML_Engineer_JD_1.pdf", "AI_ML_Engineer_JD_2.pdf", "AI_ML_Engineer_JD_3.pdf", and "Prathan_Kumar_resume.pdf".
- Chat:** The central area shows a conversation with the AI model. The user's prompt is: "Based on these three AI/ML Engineer job descriptions and my resume, predict the eight most likely interview questions that I would be asked in an AI/ML Engineer interview. Focus on both technical and behavioral questions, and ensure the questions are relevant to my projects, internships, and technical skills." The AI's response is: "Based on the requirements in the three AI/ML Engineer job descriptions and your specific background in research and systems programming, here are the eight most likely interview questions you would face."
 - Technical Questions:**
 - Generative AI Implementation (Llama 3):** All three job descriptions emphasize GenAI or advanced AI development. Because you specifically list Llama 3 in your research experience, an interviewer will likely ask: "In your UnShield research at the CCNYCS Lab, how did you specifically integrate Llama 3 into the honeypot framework, and what were the advantages of using a Large Language Model for threat intelligence over traditional methods?"
 - Reinforcement Learning Architecture:** Since you have hands-on experience with Double Deep Q-Networks (DDQN), and the job descriptions seek "advanced AI solutions", you might be asked: "Why did you select the DDQN architecture for adaptive deception strategies in your ransomware mitigation project? What were the key reward functions you designed to train the agent effectively?"
 - Feature Engineering for Security:** One job description highlights "thorough data exploration and feature engineering" as a core responsibility. When developing your threat classification models, what was your process for selecting and engineering features from cybersecurity datasets to ensure the model could accurately distinguish between benign traffic and a supply chain attack?"
 - Scaling and Production Readiness:** One job description specifically calls for "production ready quality" and "cloud or on-prem application pipelines". Given your project OS Jackfruit (building a container runtime in C), an interviewer might ask: "How does your understanding of Linux namespaces and container isolation inform how you would deploy and scale AI models in a production environment using tools like Docker?"
 - Network-AI Integration:** Your project on SDN GoS Priority Controllers demonstrates a strong grasp of networking. An interviewer might seek a bridge between these skills: "Our AI/ML Engineer role involves building scalable pipelines. How could you apply your experience with Software Defined Networking (SDN) and OpenFlow rules to optimize data throughput for a large scale AI training cluster?"
 - Behavioral Questions:**
 - Leadership and Teamwork (NCC Experience):** Every source emphasizes teamwork, communication, and collaboration. They will likely ask: "You've highlighted your National Cadet Corps (NCC) 'A' Certificate and the leadership skills you gained there. Can you describe a time when you had to use those leadership or teamwork skills to resolve a conflict within a technical project team?"
 - Troubleshooting and Algorithmic Problem Solving:** A key duty listed is "testing machine learning tools...and resolving issues with the algorithms". A likely question is: "During your research or your ISRO initiative internship, tell me about a time an AI model failed to meet its performance metrics. Walk me through your troubleshooting process and how you identified whether the issue was in the data or the algorithm itself?"
- Studio:** On the right, there is a section for creating an audio overview. It includes a "Try new Short Video Overview" button and a "Try It" button. Below this, there are several icons for different types of outputs: Audio Overview, Short Story, Video Overview, Mind Map, Reports, Podcasts, Quiz, Infographic, and Data Table.



LLM Explanation Assignment

Sources:

- Large Language Models Study Notes.pdf

Chat:

Explain the concept of Large Language Models (LLMs) in three different ways based only on the uploaded source.

- Explain it to a 12th-grade student who has never studied Artificial Intelligence before.
- Explain it to a working professional who needs the practical takeaway in under one minute.
- Explain it in one sentence to a five-year-old.

Based on the provided source, here are three ways to explain Large Language Models (LLMs):

1. For a 12th-Grade Student:

Large Language Models (LLMs) represent a major shift in how we build Artificial Intelligence. While older AI systems were "rule-based" and could only perform one specific task, LLMs are **foundation models** that can perform many different tasks—like summarizing text, translating languages, and even writing computer code—without needing to be specifically programmed for each one.

These models are built using **deep neural networks** that have been trained on **billions of words** collected from books, articles, and the internet. They work using something called the "Transformer architecture" which allows the computer to process long sequences of text and understand the context of what is being said. Essentially, the model learns to predict the next most probable word in a sentence until it generates a complete, human-like response.

2. For a Working Professional:

LLMs are advanced AI systems capable of understanding, generating, and reasoning over natural language to automate complex business processes. Unlike traditional software, they offer **multi-domain knowledge** and can be deployed for various applications—such as intelligent virtual assistants, automated content generation, and cybersecurity analysis—to **improve productivity** and handle repetitive tasks.

State-of-the-art open-source models, like **Llama 3**, provide high response accuracy and strong conversational abilities, making them powerful platforms for building intelligent software systems. However, professionals should be aware of challenges like **"hallucinations"** (incorrect information) and data bias, meaning AI-generated outputs should **always** be reviewed by a human before being used in critical applications.

3. For a Five-Year-Old:

An LLM is like a super-smart computer that has read almost every book in the world so it can

Studio:

Create an Audio Overview: [Text](#), [Audio](#), [Study Guide](#), [Mind Map](#), [Video](#)

Try new Short Video Overviews!

Audio Overview, Slide Deck, Video Overview, Mind Map, Reports, Flashcards, Quiz, Infographic, Data Table

Studio output will be saved here. After adding sources, click to add Audio Overview, Study Guide, Mind Map, and more.

AI-ML Engineer Job Ready Decoder

+ Add sources

Search the web for new sources

Web

Fast Research

AI_ML_Engineer_JD_1.pdf

AI_ML_Engineer_JD_2.pdf

AI_ML_Engineer_JD_3.pdf

Pratham_Kumar_Resume.pdf

Select all

Chat

Detailed Analysis of Findings

- Top Tier Alignment (3 of 3 JDs):** You show very strong alignment with the core requirements. Every job description emphasizes **Machine Learning, AI Development, and Data Science**. Your resume explicitly lists these in your technical skills and research intern experience. Additionally, all three descriptions value **Teamwork and Collaboration**, which you have highlighted through your NCC activities and collaborative research roles.
- Technical Strengths (2 of 3 JDs):**
 - Generative AI:** Two JDs specifically look for GenAI capabilities. Your experience with **Llama 3** and AI-driven frameworks makes you a competitive candidate here.
 - Programming: Python** is a recurring requirement, appearing both in specific job titles and general skills lists. This is your primary programming language.
 - Deployment:** The sources emphasize creating "production ready" systems. While you haven't used the specific tool **MLflow** mentioned in one JD, your experience with **Docker** and building container runtimes indicates you have the foundational skills for deployment.
- Key Missing Skill:**
 - Chatbots / Conversational AI:** While mentioned as a core use case in two job descriptions, your resume does not currently list any projects specifically focused on natural language chatbots or virtual assistants, though your work with **Llama 3** is a related foundation.
- Qualifications:** Most employers require a minimum of a **Bachelor's degree** in Computer Science or a related field. You are currently on track for this with your B.Tech at PES University, graduating in 2028. One source mentions that **certifications** like the CAIS can be preferred; while you have an NCC 'A' certificate for discipline and leadership, you currently lack a professional AI-specific certification.

Save to note

Start typing...

4 sources

+ Add sources

Search the web for new sources

Web

Fast Research

AI_ML_Engineer_JD_1.pdf

AI_ML_Engineer_JD_2.pdf

AI_ML_Engineer_JD_3.pdf

Pratham_Kumar_Resume.pdf

Select all

Today • 8:21 PM

Compare these three job descriptions and my resume. List every technical skill, programming language, framework, library, tool, certification, soft skill, and qualification that appears in at least two of the job descriptions. Rank them from most frequently mentioned to least frequently mentioned. Also indicate whether each skill is already present in my resume or is missing.

Based on the job descriptions and your resume, here is a ranked list of the technical skills, programming languages, tools, soft skills, and qualifications that appear in **at least two** of the job descriptions, along with their status in your current resume.

Skills Ranking (Most Frequently Mentioned)

Rank	Category	Skill / Qualification	Mention Frequency	Status in Resume
1	Technical	Machine Learning (ML)	3 of 3 JDs	Present
2	Technical	Artificial Intelligence (AI) Development	3 of 3 JDs	Present
3	Technical	Data Science / Data Analysis	3 of 3 JDs	Present
4	Soft Skill	Teamwork / Collaboration	3 of 3 JDs	Present
5	Language	Programming Proficiency (e.g., Python)	2 of 3 JDs	Present
6	Technical	Generative AI (GenAI)	2 of 3 JDs	Present
7	Technical	Deployment / Production	2 of 3 JDs	Present

Start typing...

4 sources

AI-ML Engineer Job Ready Decoder

+

Create notebook

Sources

+ Add sources

Search the web for new sources

Web

Fast Research

Select all

AI_ML_Engineer_JD_1.pdf

AI_ML_Engineer_JD_2.pdf

AI_ML_Engineer_JD_3.pdf

Pratham_Kumar_Resume.pdf

Chat

Based on these three AI/ML Engineer job descriptions and my resume, predict the eight most likely interview questions that I would be asked in an AI/ML Engineer interview. Focus on both technical and behavioral questions, and ensure the questions are relevant to my projects, internships, and technical skills.

Based on the requirements in the three AI/ML Engineer job descriptions and your specific background in research and systems programming, here are the eight most likely interview questions you would face.

Technical Questions

1. **Generative AI Implementation (Llama 3):** All three job descriptions emphasize **GenAI** or advanced AI development ¹ ² ³. Because you specifically list **Llama 3** in your research experience, an interviewer will likely ask: "In your UniShield research at the CCNCS Lab, how did you specifically integrate Llama 3 into the honeypot framework, and what were the advantages of using a Large Language Model for threat intelligence over traditional methods?" ⁴ ⁵

2. **Reinforcement Learning Architecture:** Since you have hands-on experience with **Double Deep Q-Networks (DDQN)** ⁴ ⁵, and the job descriptions seek "advanced AI solutions" ², you might be asked: "Why did you select the DDQN architecture for adaptive deception strategies in your ransomware mitigation project? What were the key reward functions you designed to train the agent effectively?" ⁴ ⁵

3. **Feature Engineering for Security:** One job description highlights "thorough data exploration and **feature engineering**" as a core responsibility ². You should expect: "When developing your threat classification models, what was your process for selecting and engineering features from cybersecurity datasets to ensure the model could accurately distinguish between benign traffic and a supply chain attack?" ² ³

4. **Scaling and Production Readiness:** One job description ² specifically calls for "production ready quality" and "cloud or on-prem application pipelines" ¹. Given your project **OS**

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